

Asset Scheduling Optimisation

Case Study

The Problem

A business relying on a limited fleet of shared assets to move product for multiple suppliers was losing thousands of dollars per day for each party involved. Demand for those assets was uneven and uncoordinated. Suppliers ordered trucks when they wanted them, with no visibility on what others had already booked. The result was predictable: bottlenecks, idle capacity, and scheduling decisions made under pressure and on emotion.

The Complexity

The scheduling environment involved multiple compounding variables:

Variable	Impact
Different supplier rosters	No common scheduling framework
Different load sizes and volumes	Inconsistent demand patterns
Supplier arrangements with other customers	Competing commitments outside the system
Weather and mechanical events	Unplanned lost days with no reallocation process
No forward visibility	Decisions made reactively, not proactively

The Approach

The full complexity of the scheduling environment was mapped, enabling the design of a structured solution from first principles.

Rules framework established. Clear rules defined for how suppliers access assets, what happens when a booking is missed, and what triggers reoptimisation of the asset utilisation schedule.

Schedule optimisation tool built. Full seasonal supply window projection for all suppliers. A rolling 7-day locked period with accept or decline options gave flexibility and certainty to operators now, while maintaining assurance of meeting deadlines further out.

Disruption protocol designed. When a lost day event occurs, defined rules automatically triggered reoptimisation of the full schedule. No manual intervention. No conflict.

The Impact

Metric	Before	After
Annual asset utilisation	~30%	60%+
Peak period utilisation	Untracked	Up to 80%
Individual day utilisation	Variable, often 0% or >100%	Up to 100%
Scheduling conflict	Frequent, emotionally charged	Structurally eliminated
Response to disruption	Manual, reactive, contested	Rules-based, automatic

Asset utilisation doubled. The cost of idle capacity and scheduling conflict was structurally eliminated. The complexity did not disappear. It became manageable.

What This Demonstrates

This situation required no financial data. The problem was operational with competing demands, limited capacity, no structure. The solution was built from context, rules, and first principles.

Logic in the Numbers works across financial diagnostics, decision support, and bespoke tool development. The common thread is structured thinking applied to problems that have grown complex from the inside.